

Product Returns under Perfect Competition

Draft notes

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This note records a competitive private-ownership economy with a continuum of buyers and a continuum of identical price-taking firms. The first section is the environment without returns. The second allows inspection and a refund, and the third extends that environment to two products. Unlike the incidence notes, the refund is not restricted to equal the list price.

1 No returns

1.1 Setting

The economy has a continuum of consumers, each with a numeraire endowment ω_i , and a continuum of identical price-taking firms. Consumer i 's income is ω_i plus her share of industry profit.

Each firm produces at constant unit cost c . The good cannot be returned.

Consumers have unit demand, quasi-linear preferences, and an outside option of zero. They differ in their value v_i for the good, known when they buy.

Consumers take a price p as given. Each buys at most one unit, or not at all. Buying yields $v_i - p$. She buys when $v_i \geq p$.

A firm that sells y units earns $(p - c)y$. A price-taker then compares p with c .

Industry output equals the measure of purchases.

1.2 Equilibrium

A competitive equilibrium is a price p , purchases, and industry output y such that consumers optimize given p and their income, firms maximize profit given p , and the market clears. Both sides see the same p .

Consumer i buys if and only if $v_i \geq p$. A firm that takes p as given earns $(p - c)y$, so it supplies nothing if $p < c$, any scale if $p = c$, and infinite output if $p > c$. Market clearing requires y to equal the measure of buyers. Infinite supply cannot clear against a finite mass of consumers, and zero supply cannot clear if a positive measure wants to buy. Whenever demand is positive, therefore $p = c$ and industry profit is zero. Firms then meet that demand.

2 Returns

2.1 Setting

The good can now be purchased, inspected, and returned. Checkout price and refund are two prices. Each firm still produces at cost c and now incurs a unit handling cost h on every returned item. Returned units are salvaged rather than restocked.

Consumers still have unit demand, quasi-linear risk-neutral preferences, and an outside option of zero. They differ in return hassle κ_i , the disutility of sending the good back, and in the distribution G_i of match value v realized after purchase. Consumer i knows (κ_i, G_i) when she buys, but not v .

Consumers take a checkout price p and a refund R as given. Each buys at most one unit, or not at all, then draws $v \sim G_i$ and decides whether to keep the item or return it.

If she has bought at (p, R) , keeping yields $v - p$ and returning yields $R - p - \kappa_i$. She therefore keeps if and only if $v \geq R - \kappa_i$. The keep rule depends on the refund and on hassle, not on the checkout price. Expected surplus from buying is

$$cs_i(p, R) = \mathbb{E}_{v \sim G_i} [\max(v - p, R - p - \kappa_i)] = -p + \mathbb{E}_{v \sim G_i} [\max(v, R - \kappa_i)].$$

She buys when that value is nonnegative. The reservation checkout price for a given refund, $p_i(R) = \mathbb{E}[\max(v, R - \kappa_i)]$, is the largest p at which she is willing to buy. Heterogeneity therefore cannot be collapsed into a single willingness to pay that is independent of the refund. Consumer i 's return probability at refund R is $\phi_i(R) = G_i(R - \kappa_i)$.

A firm that sells y units collects p on each sale, refunds R and incurs h on the share r that come back, and therefore earns

$$\pi = (\mu - c)y, \quad \mu = p - r(R + h),$$

where μ is the effective margin. A price-taker then compares μ with c .

Industry output equals the measure of purchases. The returned share r is the mean of $\phi_i(R)$ among those who buy, and who buys depends on (p, R) .

2.2 Equilibrium

A competitive equilibrium is a pair (p, R) , purchases, a return rate r , and industry output y such that consumers optimize given (p, R) and their income, firms maximize profit given (p, R) and r , and the market clears. Both sides see the same (p, R) .

Consumer i buys if and only if $cs_i(p, R) \geq 0$, and returns with probability $\phi_i(R)$. The return rate $r(p, R)$ is the mean of $\phi_i(R)$ among those buyers. A firm takes (p, R) and r as given and earns $(\mu - c)y$, so it supplies nothing if $\mu < c$, any scale if $\mu = c$, and infinite output if $\mu > c$.

Market clearing requires y to equal the measure of buyers, and the r in the effective margin must be $r(p, R)$. Infinite supply cannot clear against a finite mass of consumers, and zero supply cannot clear if a positive measure wants to buy. Whenever demand is positive, therefore

$$p - r(p, R)(R + h) = c. \tag{1}$$

As long as this holds, the pair is a competitive equilibrium. Consumers take (p, R) as given and follow the buy and keep rules, which determines demand and $r(p, R)$. Then $\mu = c$, so every firm is willing to sell any scale, and industry can meet that demand. Buyers optimize, sellers optimize, and the market clears.

(1) is one equation in two prices, so many pairs are competitive equilibria. The buyer's keep rule agrees with the social rule, keep when $v \geq -h - \kappa_i$, only at $R = -h$. Zero profit then requires $p = c$. That pair satisfies (1), so the efficient allocation can be sustained as a competitive equilibrium. At other pairs satisfying (1), consumers return too often or too rarely.

The First Welfare Theorem fails because the firm cannot choose the quantity of returned units. It must take back whatever the buyer returns, and its only quantity choice is how many new units to supply. Given the checkout price and the refund, the buyer determines the demand for the new good and the supply of used goods. She demands the new good when her expected surplus is

nonnegative and supplies a used unit when $v < R - \kappa_i$. The firm takes the resulting return rate as given and compares the effective margin with c .

If the firm could choose how many returned units to buy, it would gain on each return when $R < -h$ and lose when $R > -h$. The return market would therefore clear only at $R = -h$, and production would clear only at $\mu = c$. At that refund the effective margin equals the checkout price, so $\mu = c$ becomes $p = c$. The firm then earns zero on each new unit and zero on each returned unit, so it is willing to produce any quantity and to take back any quantity. Any demand for the new good and any supply of used goods can therefore clear.

3 Two products

3.1 Setting

There are now two products, indexed by $j \in \{1, 2\}$, and a separate competitive industry for each product. Industry j contains a continuum of identical price-taking firms. Its firms produce at constant unit cost c_j and incur handling cost h_j whenever a unit of product j is returned. Returned units are salvaged rather than restocked. Consumer i has numeraire endowment ω_i , together with shares of the profits earned in both industries.

Consumers are risk neutral, have quasi-linear preferences, and can buy at most one product. Before choosing, consumer i knows her return hassle κ_{ij} and the distribution G_{ij} of her match value for each product. She does not yet know either match value. If she buys product j , she then draws $v_j \sim G_{ij}$, inspects that product, and either keeps or returns it. A return ends the transaction, so she cannot use the refund to switch to the other product.

The checkout price and refund for product j are p_j and R_j . Consumers and firms take all four prices as given. If consumer i buys product j , keeping yields $v_j - p_j$, while returning yields $R_j - p_j - \kappa_{ij}$. She keeps if and only if $v_j \geq R_j - \kappa_{ij}$, and her expected surplus from choosing that product is

$$cs_{ij}(p_j, R_j) = -p_j + \mathbb{E}_{v_j \sim G_{ij}} [\max(v_j, R_j - \kappa_{ij})].$$

She chooses the product with the larger nonnegative expected surplus and buys neither when both surpluses are negative. Assume ties are resolved by a fixed rule. Conditional on buying product j , her return probability is

$$\phi_{ij}(R_j) = G_{ij}(R_j - \kappa_{ij}).$$

Let $B_j(\mathbf{p}, \mathbf{R})$ be the set of consumers who choose product j at checkout prices $\mathbf{p} = (p_1, p_2)$ and refunds $\mathbf{R} = (R_1, R_2)$. Industry demand D_j is the measure of this set. When $D_j > 0$, its return rate is the mean return probability among those buyers,

$$r_j(\mathbf{p}, \mathbf{R}) = \frac{1}{D_j} \int_{B_j(\mathbf{p}, \mathbf{R})} \phi_{ij}(R_j) di.$$

Although a buyer's keep decision depends only on the refund for the product she chose, each return rate depends on all four prices because those prices determine who selects each product.

A firm in industry j takes (p_j, R_j) and the market return rate r_j as given. If it sells q_j units, it earns

$$\pi_j = (\mu_j - c_j)q_j, \quad \mu_j = p_j - r_j(R_j + h_j).$$

The firm chooses only its scale. The return rate is not another price or a choice of the firm. It is determined by consumers and must agree with the market return rate that firms take as given. Industry output for product j equals D_j .